DISCRETE MATH · THEORY OF GRAPHS AND TREES

Graph Explorer: Euler, Hamilton & Trees

Explore & Reflect **TEACHER KEY**

Name _____ Date _____ Period _____

Open **Graph Explorer: Euler, Hamilton & Trees** in your unit's Interactive Math section on Mr. K's Classroom (or use your teacher's link). Answer Part 1 **before** you use the tool. Use the tool for Parts 2 and 3. Do Parts 4 and 5 on your own.

1 Predict before you touch the tool

1. A graph has degrees 3, 3, 2, 2, 2. How many edges does it have?
   (3 + 3 + 2 + 2 + 2)/2 = 6.
2. Can you draw the Königsberg bridges (degrees 5, 3, 3, 3) without lifting your pencil or repeating a bridge? Predict and explain.
   No: an Euler trail allows at most two odd-degree vertices, and all four are odd.

2 Explore use the tool

1. Load Königsberg. Add one bridge so an Euler trail exists. Between which land masses must it go?
   Between any two of the four odd vertices (for example A and C); then exactly two vertices are odd.
2. Load the Bowtie. Does it have an Euler circuit? A Hamiltonian cycle? Why?
   Euler circuit yes (all degrees even); no Hamiltonian cycle (the middle vertex would be visited twice).
3. Load C_5 and turn on Coloring. What is χ? Is there a triangle?
   $\chi = 3$ with no triangle: an odd cycle needs 3 colors.
4. Load the weighted road network and run Kruskal. Record the edges taken, the edge skipped and the total.
   Take AC 2, CD 3, AB 4, DE 6; skip BC 5 (cycle); stop. Total 15.

3 Record use the tool and fill in the table

Graph facts

Load each graph and read the facts panel.

Graph	Vertices	Edges	Euler	Hamiltonian cycle?	χ
K_4	4	6	none (4 odd)	yes	4
K_5	5	10	circuit	yes	5
$K_{3,3}$	6	9	none (6 odd)	yes	2
Petersen	10	15	none (10 odd)	no	3
House	5	6	trail	yes	3

4 Explain what did you discover?

1. The sum of the degrees equals 2 times the number of edges.
2. A connected graph has an Euler circuit exactly when every degree is even, and an Euler trail (not circuit) when exactly two degrees are odd.

3. A connected graph with n vertices is a tree exactly when it has n – 1 edges.

4. Explain the difference between an Euler circuit and a Hamiltonian cycle, and why there is a simple test for one but not the other.
   An Euler circuit uses every edge once; a Hamiltonian cycle visits every vertex once. Euler circuits are decided by degrees (connected, all even); no simple condition decides Hamiltonian cycles, so in general you must search.

5 Apply a new situation, no tool

1. A tree has 12 vertices. How many edges? Can it have a vertex of degree 12?
   11 edges. No: a vertex can be adjacent to at most the other 11.

2. Is there a simple graph with degrees 3, 3, 3, 1? Explain.
   No: the sum 10 is even, but three vertices of degree 3 would each need to be adjacent to all others, so the last vertex would have degree 3, not 1.

3. Exams A, B, C, D, E: pairs sharing a student are AB, AC, BC, BD, CD, DE. What is the fewest number of time slots?
   3 (triangle ABC forces 3; for example A, D in slot 1; B, E in slot 2; C in slot 3).

4. K_6: how many edges? Euler circuit?
   15 edges; every degree is 5 (odd), so no Euler circuit or trail.

Teacher notes

Intended discovery: Degree sum is 2|E|; Euler circuits and trails are decided by connectivity and odd degrees; Hamiltonian cycles need a search; trees have n – 1 edges; $\chi \geq 3$ for any odd cycle; Kruskal builds an MST by skipping cycle-forming edges.

Likely misconceptions:

- Confusing Euler (edges) with Hamilton (vertices).
- Thinking a triangle is the only reason for 3 colors.
- Counting a loop as adding 1 to the degree.

Answers that can vary: Explore 1: any edge joining two odd vertices; Apply 3 colorings vary.